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The impact of thermodynamic forcing on cloud microphysics studied by direct numerical simulations

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Numerical representation of atmospheric clouds is challenging when entrainment and mixing are strong. Because both velocity and thermodynamic fluctuations need to be sustained through external forcing. Although momentum forcing is common in direct numerical simulations (DNS) of clouds, thermodynamic forcing has not gained enough attention. We adopt a particle-based DNS approach in this study and implement forcing in the momentum, temperature, and vapor mixing ratio fields. All forcing terms are constructed in the Fourier space using a band of low wavenumbers. The turbulent flow of air is assumed to be homogeneous and isotropic. Aerosol particles and cloud droplets are tracked in Lagrangian fashion. Our initial setup is such that the cloud and environment are separated by a sharp interface and subsequent turbulent mixing is simulated. The results suggest that thermodynamic forcing sustains thermodynamic fluctuations, drives the mean thermodynamic fields to equilibrium faster and suppresses microscale phase change. The probability distributions of thermodynamic fields are highly non-Gaussian if thermodynamic forcing acts in random directions. Strong thermodynamic forcing can create a locally supersaturated condition even though the environment is subsaturated, allowing some droplets to grow while most of them shrink. It is found that thermodynamic forcing broadens the size distributions of particles, prevents the decay of relative dispersions of particle radii and maintain a balance turbulent mixing and phase change.

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NOMENCLATURE

c_p, C_d, Da	= specific heat of air [J/(kg.K)], rate of mass exchange [1/s], Damkohler number [-]
δ_{k,k_f}, δ_T, δ_{q_v}	= delta functions [-] in the Fourier space, temperature field, and vapor field, respectively
$\epsilon_{T,in}$, $\epsilon_{q_v,in}$	= dissipation rates in the initial temperature [K^2/s], and vapor [$(g/kg)^2$/s] fields
ϵ, η	= turbulent kinetic energy dissipation rate [m^2/s^3], Kolmogorov scale [m]
$\boldsymbol{F}_b$, $\boldsymbol{F}_c$	= buoyancy force [m/s^2], external force [m/s^2] in the momentum field per unit mass
F_T, F_{q_v}	= external forcing in the temperature [K/s], water vapor [(g/kg)/s] fields
$\boldsymbol{g}$, $\boldsymbol{k}_f$	= acceleration vector due to gravity [m/s^2], forcing wavenumber vector [1/m]
k_{min}, k_{max}	= minimum wavenumber [1/m], maximum wavenumber [1/m]
κ, k_T	= hygroscopicity of solute [-], thermal conductivity in air [W/(m.K)]
l_0, l	= length scale characterizing large eddies [m], integral length scale [m]
L_h, λ, M_l	= specific latent heat [J/kg], Taylor microscale [m], molar mass of water [kg/mol]
m_a, μ_T, μ_v	= mass of air [kg], molecular diffusivities [m^2/s] of temperature, water vapor in air
ν, p	= kinematic viscosity of air [m^2/s], instantaneous fluid pressure [N/m^2]
q_v, $q_{v,s}$	= instantaneous water vapor mixing ratio [g/kg], saturation vapor mixing ratio [g/kg]
r_d, r	= dry aerosol radius [μm], wet radius of aerosol particle or cloud droplet [μm]
r_c, R_v	= critical radius [μm], specific gas constant for water vapor [J/(kg.K)]
$R_{q_v' q_v'}$	= auto-correlation coefficient between vapor mixing ratio fluctuations [-]
ρ_a, ρ_l, S_e	= density of air [kg/m^3], density of water [kg/m^3], environmental supersaturation [-]
S_k, S_c	= particle equilibrium supersaturation [-], critical equilibrium supersaturation [-]
Sc, S_t, σ_l	= Schmidt number [-], Stokes number [-], surface tension of water [N/m]
T, τ_p	= instantaneous temperature [K], particle response time [s]
τ_t, τ_c	= turbulent mixing timescale [s], phase change timescale [s]
$\boldsymbol{u}$, u_{rms}	= instantaneous fluid velocity vector [m/s], root-mean-square velocity [m/s]
$\boldsymbol{V}$, $\boldsymbol{X}$	= particle velocity vector [m/s], particle position vector [m]
PDF, TKE	= probability density function [-], turbulent kinetic energy [m^2/s^3]

I. INTRODUCTION

Numerical modeling has become a powerful tool to investigate turbulence-cloud-aerosol interactions in the atmosphere as experimental measurements are few and difficult to conduct. The most accurate approach is direct numerical simulation (DNS) coupled with Lagrangian particle tracking. This method resolves the flow of turbulent air upto the smallest scale without any kind of modeling of turbulent eddies and tracks cloud droplets and aerosol particles individually in Lagrangian fashion. The thermodynamic (scalar) fields of temperature and vapor mixing ratio are transported in Eulerian framework. Flow turbulence is assumed to be homogeneous and isotropic, but it decays in absence of a turbulence production mechanism. To prevent the decay, many DNS studies (Vaillancourt,[1] Kumar et al.,[2] Li[3]) and our previous DNS model (Gao et al.,[4] Sharfuddin et al.[5]) applied external forcing in the momentum field. Although velocity turbulence was sustained, thermodynamic fluctuations decayed in those models. In the atmosphere, thermodynamic fluctuations are sustained by strong entrainment and mixing in the cloud edge[6,7] while they decay in the cloud core. To numerically represent entrainment and mixing in the edge of clouds, thermodynamic fields must be forced besides momentum forcing. The objective of present DNS study is to simulate turbulence-cloud-aerosol interactions with desired levels of thermodynamic fluctuations.

The most common forcing method is Fourier forcing which is carried out in a band of low wavenumbers in the Fourier space. The idea is to inject energy at larger scales which cascades down to smaller scales. Eswaran and Pope[8] showed that Fourier forcing of the momentum field results in a statistically stationary velocity field. Bogucki et al.[9] extended the concept of Fourier forcing to a passive scalar (thermodynamic) field. Watanabe and Gotoh[10] applied Fourier forcing in the momentum and scalar fields. They suggested that the scaling behavior of a passive scalar differs from that of velocity. Chen and Cao[11] applied Fourier forcing in the scalar field and found that scalar fluctuations are highly intermittent at small scales. Also, they become decorrelated more rapidly than velocity fluctuations do. In the context of atmospheric clouds, the DNS model of Paoli and Shariff[12] implemented Fourier forcing in the momentum, temperature, and water vapor mixing ratio fields. They reported that temperature and vapor forcings broaden the size distributions of cloud droplets. The DNS model of Chen et al.[13] forced the thermodynamic fields in a low wavenumber band to simulate the growth of mixed-phase clouds. Satio et al.[14] applied Fourier forcing in the water vapor field but kept the temperature field unforced. Their DNS study showed that external vapor forcing can maintain fluctuations in the supersaturation as required.

Large-scale forcing drives the evolution of the total (mean plus fluctuating) thermodynamic fields. The cloud-resolving 2D model of Fu et al.[15] included temperature and vapor forcing terms which were constructed by multiplying the mean vertical velocity with the vertical gradients of mean temperature and water vapor, respectively. This kind of forcing is not suitable for DNS as the mean velocity is zero in homogeneous and isotropic turbulence (HIT). Moreover, Fu et al.[15] derived the mean gradients from experimental observations which cannot be obtained easily at the scales resolved by DNS. Shen et al.[16] also used the mean vertical velocity to force the temperature and water vapor fields in a cloud-resolving model that had a much larger domain size but coarser grid resolution than typical DNS studies. They reported that large-scale forcing in the upward direction increases the quantity of water vapor but does the opposite in the downward direction. Thomas et al.[17] applied external forcing in the vapor and temperature fields in their large eddy simulations (LES) of turbulent clouds. The authors suggested that thermodynamic forcing is required to account for the entrainment of ambient air into the cloud. Li and Shen[18] varied the magnitude of vapor forcing and concluded that cloud microphysical response changes accordingly. Tao et al.[19] found that large-scale forcing affects the domain-averaged temperature and water vapor while producing a warmer and more humid thermodynamic state.

Knowledge gaps still exist regarding the impact of thermodynamic forcing on cloud physics. Specifically, we need improved understanding of microscale cloud processes such as aerosol activation, condensation, cloud droplet deactivation, and evaporation if thermodynamic fluctuations persist. Hence, we extend our previous DNS model[5] to include external forcing in the temperature and vapor mixing ratio fields. The forcing terms are constructed in the Fourier space at the large scales or low wavenumbers.[8] The initial setup consists of the cloudy and clear-air regions separated by a sharp interface. We simulate the entrainment of environmental air into the cloud volume and subsequent turbulent mixing. The initial thermodynamic fields change abruptly across the interface in our model unlike those in Paoli and Shariff[12] and Chen et al.[13] The results are compared for external forcing in the temperature and vapor fields, the vapor field only, and no thermodynamic fields. Also, the magnitude of external forcing is varied in the momentum and thermodynamic fields. The evolutions of the mean and fluctuating thermodynamic fields along with corresponding cloud microphysical responses are thoroughly analyzed.

II. GOVERNING EQUATIONS

The equations solved in our model are described in this section. They include the Navier-Stokes equations, the transport equations for thermodynamic variables, the equations for external forcings, the equations for kappa-Kohler theory, and Lagrangian particle tracking equations.

A. Turbulent velocity field and momentum forcing

The turbulent for of air is assumed to be governed by the Navier-Stokes equations:

$$\frac{\partial \boldsymbol{u}}{\partial t} + (\boldsymbol{u}.\nabla)\boldsymbol{u} = -\frac{1}{\rho_a}\nabla p + \nu\nabla^2\boldsymbol{u} + \boldsymbol{F}_b + \boldsymbol{F}_c\,, \tag{1}$$

$$\nabla.\boldsymbol{u} = 0\,. \tag{2}$$

Here, $\boldsymbol{u}$ is the instantaneous fluid (air) velocity vector and p is the instantaneous fluid pressure, $\boldsymbol{F}_b$ is the buoyancy force, and $\boldsymbol{F}_c$ is the external force applied to sustain flow turbulence. The $\boldsymbol{F}_b$ can be expressed as:[20]

$$\boldsymbol{F}_b = -\boldsymbol{g}\left[\frac{T-\langle T\rangle}{\langle T\rangle} + 0.608(q_v - \langle q_v\rangle) - q_c\right]. \tag{3}$$

where T is the instantaneous temperature, q_v and q_c are the instantaneous water vapor mixing ratio and liquid water mixing ratio,[5] respectively. $\langle T\rangle$ and $\langle q_v\rangle$ are the domain averages of T and q_v, respectively. Our model's domain size (0.512 m) is much smaller compared to an actual cloud and thus $\boldsymbol{F}_b$ is found to be negligible. Since $\boldsymbol{F}_b$ cannot prevent a complete decay of the imposed initial turbulent field, momentum forcing is added in Eq. (1). This external forcing is formulated in the Fourier space as:[21]

$$\hat{\boldsymbol{f}}_c(\boldsymbol{k},t) = \epsilon_{in}\frac{\hat{\boldsymbol{u}}(\boldsymbol{k},t)}{\sum_{k_f \epsilon_k}\left|\hat{\boldsymbol{u}}(\boldsymbol{k}_f,t)\right|^2}\delta_{k,k_f}\,. \tag{4}$$

where ϵ_{in} is an input TKE dissipation rate, $\boldsymbol{k}$ is the wavenumber vector, $\hat{\boldsymbol{u}}(\boldsymbol{k},t)$ is the Fourier-transformed velocity field, $\boldsymbol{k}_f$ is the forced wavenumber vector, and δ_{k,k_f} is a delta function that is unity inside the wavenumber shell consisting of the forced wavenumbers but zero otherwise. The $\hat{\boldsymbol{f}}_c$ is inverse transformed into the physical space as $\boldsymbol{F}_c$.

B. Transport and forcing of the thermodynamic fields

The transport equations for temperature (T) and water vapor mixing ratio (q_v) are written as:

$$\frac{\partial T}{\partial t} + (\boldsymbol{u}.\nabla)T = \frac{L_h}{c_p}C_d + \mu_T\nabla^2 T + F_T\,, \tag{5}$$

$$\frac{\partial q_v}{\partial t} + (\boldsymbol{u}.\nabla)q_v = -C_d + \mu_v\nabla^2 q_v + F_{q_v}\,. \tag{6}$$

where C_d quantifies the time rate of mass exchange between liquid water and water vapor and is described by Eq. (19). L_hC_d/c_p and $-C_d$ are the latent heat terms in the respective equations. μ_T and μ_v are the molecular diffusivities of temperature and water vapor, respectively. F_T and F_{q_v} are the external forcing terms for T and q_v, respectively. Let

θ be a placeholder for T and q_v. Similar to Equation (4), thermodynamic forcing can be constructed in the Fourier space as:

$$\hat{f}_\theta(\boldsymbol{k},t) = \epsilon_{\theta,in} \frac{\hat{\theta}(\boldsymbol{k},t)}{\sum_{k_f \epsilon_k} \left|\hat{\theta}(\boldsymbol{k}_f,t)\right|^2} \delta_{k,k_f} \,. \tag{7}$$

Above, $\hat{\theta}(\boldsymbol{k},t)$ is the Fourier-transformed thermodynamic field and $\epsilon_{\theta,in}$ is the dissipation rate in the thermodynamic field at initial time. The $\hat{f}_\theta(\boldsymbol{k},t)$ is converted back to the physical space and becomes F_θ (F_T or F_{q_v}). Note that the evolution of T and q_v depends on the direction of latent heat change. For example, C_d is positive during condensation and thus T increases while q_v decreases. The opposite holds true during evaporation. With inclusion of F_θ, interaction between external forcing and latent heat is expected as both terms have significant magnitude. The sign of F_θ is random based on Eq. (7) and does not necessarily match the signs of the latent heat (phase change) terms. To better understand the impact of F_θ on the evolution of T and q_v, we adjust the direction of F_θ. First, we let F_θ act in random directions, for which Equations (5) and (6) are applicable as they are. Second, we impose the condition that F_θ always act in the direction of latent heat change by introducing two delta functions δ_T and δ_{q_v} defined as:

$$\delta_T = \begin{cases} 1 & if \ \dfrac{L_h}{c_p} C_d > 0 \\ -1 & if \ \dfrac{L_h}{c_p} C_d < 0 \end{cases} \quad , \qquad \delta_{q_v} = \begin{cases} 1 & if -C_d > 0 \\ -1 & if -C_d < 0 \end{cases} . \tag{8}$$

With the foregoing, F_T and F_{q_v} are written as:

$$F_T = \delta_T |F_T| \ , \qquad F_{q_v} = \delta_{q_v} \left|F_{q_v}\right| . \tag{9}$$

A third setting is possible in which F_θ always acts in the direction opposite to the latent heat change. But we find that to be unphysical as explained in Appendix A.

C. Growth of cloud droplets and aerosol particles

The motion of each aerosol particle or cloud droplet is represented in the Lagrangian fashion. The simplified expressions for the Lagrangian position and velocity are[2]

$$\frac{d\boldsymbol{X}_i(t)}{dt} = \boldsymbol{V}_i(t) \,, \tag{10}$$

$$\frac{d\boldsymbol{V}_i(t)}{dt} = \frac{1}{\left(\tau_p\right)_i} [\boldsymbol{u}_i(\boldsymbol{X},t) - \boldsymbol{V}_i(t)] + \boldsymbol{g} \,. \tag{11}$$

where the subscript 'i' denotes the i-th aerosol particle or cloud droplet, $\boldsymbol{V}$ is its velocity vector, and $\boldsymbol{X}$ is its position vector. $\boldsymbol{u}_i$ is the instantaneous fluid velocity vector at particle position i and is obtained through a trilinear interpolation of the Eulerian field. The particle response time τ_p is given by[2]

$$\left(\tau_p\right)_i = \frac{2\rho_l r_i^2}{9\rho_a \nu}. \tag{12}$$

Equation (10) is valid when the particle Stokes number (S_t) is significantly less than unity. The time evolution of particle radius (r) is described by[22]

$$\frac{dr_i(t)}{dt} = \frac{G_i}{r_i(t)}\left(S_e(\boldsymbol{X},t) - S_k(\boldsymbol{X},t)\right)_i. \tag{13}$$

where S_e is the environmental supersaturation and S_k is the particle equilibrium supersaturation that accounts for the curvature and solute effects combined. The kappa-Kohler theory[23] expresses the S_k as:

$$S_k = \frac{A}{r} - \frac{\kappa r_d^3}{r^3 - r_d^3(1-\kappa)}, \qquad A = \frac{2\sigma_l M_l}{RT\rho_l}. \tag{14}$$

where r_d is the dry aerosol radius, κ is the hygroscopicity of solute (dry aerosol), R is the universal gas constant, σ_l is the surface tension of water, and M_l is molar mass of water. If the atmosphere is supersaturated, aerosol particles grow beyond the critical radii and activate into cloud droplets. Whereas cloud droplets shrink below the critical radii and deactivate into aerosol particles in a subsaturated environment. Using the kappa-Kohler theory, the critical radius (r_c) and critical equilibrium supersaturation (S_c) are derived as:[24]

$$r_c = \sqrt{\frac{3\kappa r_d^3}{A}}, \qquad S_c = \frac{2}{\sqrt{\kappa}}\left(\frac{A}{3r_d}\right)^{\frac{3}{2}}. \tag{15}$$

A particle can be classified as aerosol particle or cloud droplet by comparing r with r_c. An aerosol particle corresponds to $r_d < r \leq r_c$ while a cloud droplet corresponds to $r > r_c$. The S_e depends on the q_v and T and is expressed as:

$$S_e = \frac{q_v}{q_{v,s}} - 1, \quad q_{v,s} = 621.97\frac{p_{sat}}{p - p_{sat}}, \tag{16}$$

$$p_{sat} = 611.2 \times \exp\left(\frac{17.67(T - 273.15)}{T - 29.65}\right). \tag{17}$$

$q_{v,s}$ is the saturation vapor mixing ratio[25] and p_{sat} is the saturation vapor pressure.[26] The growth parameter G is found as:[22]

$$G = \frac{1}{\frac{L_h \rho_l}{k_T T}\left(\frac{L_h}{R_v T} - 1\right)(1 + S_k) + \frac{\rho_l R_v T}{\mu_v p_{sat}}}, \tag{18}$$

where k_T is the thermal conductivity in air and R_v is the specific gas constant for water vapor. The C_d is determined as:

$$C_d(\boldsymbol{x}, t) = \frac{4\pi\rho_l}{\rho_a a^3}\sum_{i=1}^{n} G_i r_i(t)\big(S_e(\boldsymbol{X}, t) - S_k(\boldsymbol{X}, t)\big)_i . \tag{19}$$

where n is the number of particles in a grid cell that has a volume of a^3.

III. NUMERICAL DETAILS

We run simulations for two scenarios: Scenario I and Scenario II.[5] The first is for condensation and activation of aerosol particles into cloud droplets, while the second is for evaporation and deactivation of cloud droplets into aerosol particles. Since the domain size for this study is much smaller than that of a natural cloud, it will be difficult to observe significant condensation and evaporation at the same time in our model. So, we design two separate scenarios that correspond to different initial conditions. In the first scenario, the cloud volume mixes with saturated ambient air, and condensation prevails. In the second, the dry (subsaturated) air is entrained into the cloud volume and evaporation is dominant.

A. Momentum, thermodynamic, and particle fields

We vary the range of forced wavenumbers to generate two levels of flow turbulence: 'low' and 'high' and denote them as 'L' and 'H', respectively.[5] Momentum forcing is applied in the wavenumber bands: $1 \le k_f/k_{min} \le 4.12$ and $1 \le k_f/k_{min} \le 16.18$, respectively. The minimum wavenumber $k_{min} = 2\pi/L = 2\pi/0.512 \sim 12.27$, the maximum wavenumber $k_{max} \sim 128 k_{min}$, and the initial TKE spectrum is specified as[27]

$$E(k) = \frac{16}{\sqrt{\pi/2}} \frac{u_0^2 k^4}{k_0^5} \exp\left(-\frac{2k^2}{k_0^2}\right). \tag{20}$$

where $k_0 = 4.12 k_{min}$ and $u_0 = 0.16\ m/s$ is the root-mean-square (rms) velocity. Table I lists the turbulence statistics such as the turbulent kinetic energy, $\hbar$, its dissipation rate, ϵ, the rms velocity, u_{rms}, the Kolmogorov length scale η, the Taylor micro-scale, λ, the length scale characterizing large eddies, l_0, the forcing Reynolds number, Re_f, the Taylor-scale Reynolds number, Re_λ, and the Reynolds number based on l_0, Re_{l_0}. The definitions of abovementioned quantities can be found in Ref. 28.

Table I. Statistics of the turbulent velocity field at 4 s

Turbulence level	$\hbar$ (m^2/s^2)	ϵ (m^2/s^3)	u_{rms} (m/s)	λ (m)	η (m)	l_0 (m)	Re_f	Re_λ	Re_{l_0}
High	0.0094	0.00494	0.1386	0.0169	0.00091	0.184	401.12	156.11	1704
Low	0.0025	0.00055	0.0797	0.0261	0.00157	0.227	192.97	138.74	1207

Our model resembles a 3D cube inside which flow turbulence, cloud droplets, and aerosol particles interact. The boundary conditions are triply periodic, and the domain volume is $0.512^3\ m^3$. At the start of simulations, the domain is divided equally into cloudy and clear-air regions. A slab-like configuration[4] is assumed for the initial field (Fig. 1), to enable a sharp transition in the water vapor mixing ratio and temperature at the cloud/clear-air interface. Such an inhomogeneous distribution of the initial thermodynamic fields allows us to realistically simulate turbulent mixing between the cloud and environment.[29]

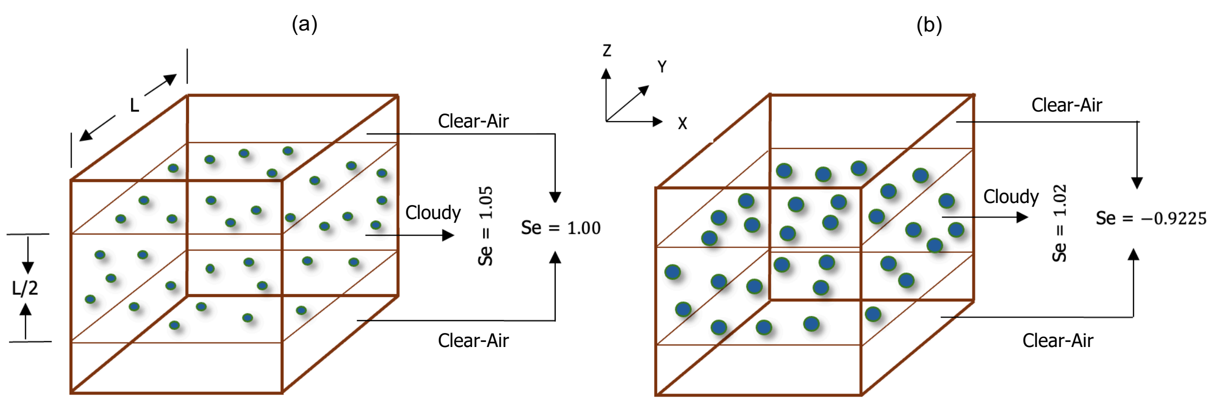

FIG. 1 Initial configurations in (a) Scenario I, and (b) Scenario II. The middle half represents the cloudy region while the top and bottom one-fourths constitute the clear-air region. A sharp interface separates the two regions. The cloudy region is initially 5% supersaturated in Scenario I and 2% in Scenario II. The clear-air region is just saturated in the first and 92.25% subsaturated in the second. The spheres represent dry aerosols in Scenario I and cloud droplets in Scenario II.

In Scenario I, the cloudy region is supersaturated (humid), but the clear-air region is just saturated at initial time. The initial vapor mixing ratio field is specified as:[4]

$$q_v(z, t = 0) = \begin{cases} q_v^{max}, & d - L \times 0.5 \times 0.5 \leq z < d + L \times 0.5 \times 0.5 \\ q_{v,e}, & elsewhere \end{cases}. \quad (21)$$

Above, $q_v^{max} = 4.066\ g/kg$, and $q_{v,e} = 3.872\ g/kg$. The length, $d \equiv L/2$ is the width of the cloud slab. q_v^{max} and $q_{v,e}$ belong to the cloudy and clear-air regions, respectively. The temperature field is initialized as:[4]

$$T(t = 0) = \langle T(t = 0)\rangle - 0.608\langle T(t = 0)\rangle(q_v(z, t = 0) - \langle q_v(t = 0)\rangle)\,. \quad (22)$$

To resolve the discontinuous distribution (Eqs. (21) and (22)), we choose a finite difference based DNS model[5] instead of a pseudospectral DNS model because the latter could give rise to artificial oscillations (Gibbs phenomenon).[2] We simulate seven cases in Scenario I which are summarized in Table II. The identifiers 'L' and 'H' before the dash (-) imply 'low' and 'high' levels of flow turbulence, respectively. The identifiers 'L' and 'H' after the dash (-) indicate 'low' and 'high' levels of thermodynamic forcing with forcing wavenumber ranges being $1 \leq k_f/k_{min} \leq 4.12$ and $1 \leq k_f/k_{min} \leq 16.18$, respectively. The identifier 'LR' suggests 'low' level of thermodynamic forcing that acts in 'random' directions while 'LV' implies 'low' level of forcing applied in the 'vapor' field only. At the start, 16×10^6 dry aerosols are randomly placed in the cloudy region, giving a number concentration of 119 cm^{-3}. The size (radius) distributions of dry aerosols are assumed to be lognormal with a geometric mean of 0.126 μm and a standard deviation of 0.02 μm. Dry aerosols absorb water from the environment to become aqueous aerosol particles which grow further and activate into cloud droplets in Scenario I.

TABLE II. Cases in Scenario I and Scenario II.

Case	Dry aerosol size (μm)	TKE dissipation rate (m^2/s^3)	Momentum forcing level	Level, field of thermodynamic forcing (F_θ)	Directionality of F_θ
L	0.126 ± 0.02	0.00055	Low	N/A	N/A
L-L	0.126 ± 0.02	0.00055	Low	Low, both	Same
L-H	0.126 ± 0.02	0.00055	Low	High, both	Same
L-LR	0.126 ± 0.02	0.00055	Low	Low, both	Random
L-LV	0.126 ± 0.02	0.00055	Low	Low, vapor only	Same
H	0.126 ± 0.02	0.00494	High	N/A	N/A
H-L	0.126 ± 0.02	0.00494	High	Low, both	Same

Case	Comments
L	Level of momentum forcing is low (L). Thermodynamic forcing is absent or $F_T = F_{q_v} = 0$.
L-L	Level of momentum forcing is low (L). Magnitudes of F_T and F_{q_v} are low (L). F_T and F_{q_v} act in the directions of latent heat change.
L-H	Level of momentum forcing is low (L). Magnitudes of F_T and F_{q_v} are high (H). F_T and F_{q_v} act in the directions of latent heat change.
L-LR	Level of momentum forcing is low (L). Magnitudes of F_T and F_{q_v} are low (L). F_T and F_{q_v} act in random directions.
L-LV	Level of momentum forcing is low (L). Magnitude of F_{q_v} is low (L). Only the vapor (V) field is forced and $F_T = 0$. F_{q_v} acts in the direction of latent heat change.
H	Level of momentum forcing is high (H). Thermodynamic forcing is absent or $F_T = F_{q_v} = 0$.

H-L	Level of momentum forcing is high (H). Magnitudes of F_T and F_{q_v} are low (L). F_T and F_{q_v} act in the directions of latent heat change.

The cloudy region is supersaturated (humid) but the clear-air region is subsaturated (dry) in Scenario II at initial time with $q_v^{max} = 3.949\ g/kg$ and $q_{v,e} = 0.30\ g/kg$. A total of 16×10^6 cloud droplets, each of them having a radius (r) of 10 μm, are randomly placed in the cloudy region at the start. Dry aerosols, also known as solutes, are assumed to be present in the core of cloud droplets. With r_d known, r_c can be calculated using Eq. (15). The dry and critical radii are fixed at each particle position but the wet radii (r) evolve with time. We again simulate seven cases (Table II) in Scenario II as cloud droplets evaporate and deactivate into aerosol particles.

B. Parameters, grid resolution, and computing resources

The constants and parameters used in our model are listed in Table III. They include the domain-averaged temperature, pressure, and Stokes number (S_t) at initial time. The dry aerosol composition is ammonium sulfate for which the value of hygroscopicity (κ) is 0.61.[30] The TKE dissipation rate becomes statistically stationary by 2 s after which particles are allowed to grow (Eq. (13)). The dissipation rates in the initial temperature ($\epsilon_{T,in}$) and vapor mixing ratio ($\epsilon_{q_v,in}$) fields are shown in Table III. The Schmidt number ($Sc = \nu/\mu_T$) is 0.68 for both temperature and vapor mixing ratio. But the effective Sc is higher because thermodynamic fields are coupled with particles in our model. Note that the mean temperature, pressure, and density do not vary with height in our microscale model. A detailed analysis in this regard can be found in Ref. 5.

TABLE III. Parameters and constants

Symbol	Value	Unit	Symbol	Value	Unit
q_v^{max}	4.066 (Scenario I) 3.949 (Scenario II)	g/kg	$q_{v,e}$	3.872 (Scenario I) 0.30 (Scenario II)	g/kg
L_h	2.5×10^6	J/kg	c_p	1005.0	J/kg/K
ρ_a	0.92	kg/m^3	ν	1.5×10^{-5}	m^2/s
ρ_l	1000	kg/m^3	$\langle p(t=0) \rangle$	71446	N/m^2
$\langle T(t=0) \rangle$	270.75	K	R_v	461.5	J/kg/K
k_T	0.0238	W/m/K	σ_l	0.072	N/m
$\mu_T = \mu_v$	2.2×10^{-5}	m^2/s	M_l	0.018	kg/mol
R	8.314	J/mol/K	$\langle S_t(t=0) \rangle$	7.35×10^{-8} (Scenario I) 4.63×10^{-4} (Scenario II)	-
κ	0.61	-			

$\epsilon_{T,in}$	2.16×10^{-5} (Scenario I) 7.7×10^{-4} (Scenario II)	K^2/s	$\epsilon_{q_v,in}$	7.9×10^{-6} (Scenario I) 2.8×10^{-3} (Scenario II)	$(g/kg)^2/s$

To balance the required grid resolution and computational cost, we use 256^3 grid points for all simulations. The $k_{max}\eta$ values for 'low' and 'high' turbulence levels are 2.46 and 1.43, respectively. Our DNS model has been built using the FronTier software package.[31] We utilize the Perlmutter supercomputer managed by the National Energy Research Scientific Computing Center (NERSC), United States Department of Energy, USA to run simulations.

IV. RESULTS AND DISCUSSIONS

We discuss the impact of external forcing on the fluctuating and mean thermodynamic fields in Sec. IV A. Then, we analyze the impact of thermodynamic forcing on microscale condensation, aerosol activation, evaporation, and cloud droplet deactivation in Sec. IV B. This is followed by Sec. IV C that explains the relationship between external forcing and turbulent mixing. The concluding remarks in Sec. V summarize our findings.

A. The impact of external forcing on thermodynamic fields

1. Condensational growth (Scenario I)

The temporal evolution of the variances of temperature (σ_T^2), vapor mixing ratio ($\sigma_{q_v}^2$), and supersaturation ($\sigma_{S_e}^2$) is shown in Figs. 2(a)-(c), respectively. The plots represent Cases L, L-L, L-H, L-LR, and L-LV in Scenario I which simulates condensation and aerosol activation. We see that thermodynamic fluctuations attain statistical stationarity when forcing is applied but decay otherwise. The thermodynamic fields are inhomogeneously distributed at initial time in our model. As simulations progress, turbulent mixing homogenizes the thermodynamic fields and fluctuations diminish (Case L). On the other hand, forcing acts as a source of fluctuations and prevents the decay. We find that σ_T^2, $\sigma_{q_v}^2$, and $\sigma_{S_e}^2$ have a smooth profile when forcing direction is random (Case L-LR) compared to the other forced cases (Cases L-L, L-H, and L-LV). When statistical stationarity is reached, the σ_T^2 and $\sigma_{q_v}^2$ are larger for Case L-H compared to Case L-L as the magnitude of thermodynamic forcing is higher for the former. The σ_T^2 decays for Case L-LV since the temperature field is unforced and the decay is not monotonic. The profiles of $\sigma_{S_e}^2$ are similar for Cases L-L and L-LV which suggests that vapor forcing influences supersaturation fluctuations more strongly than temperature forcing does.

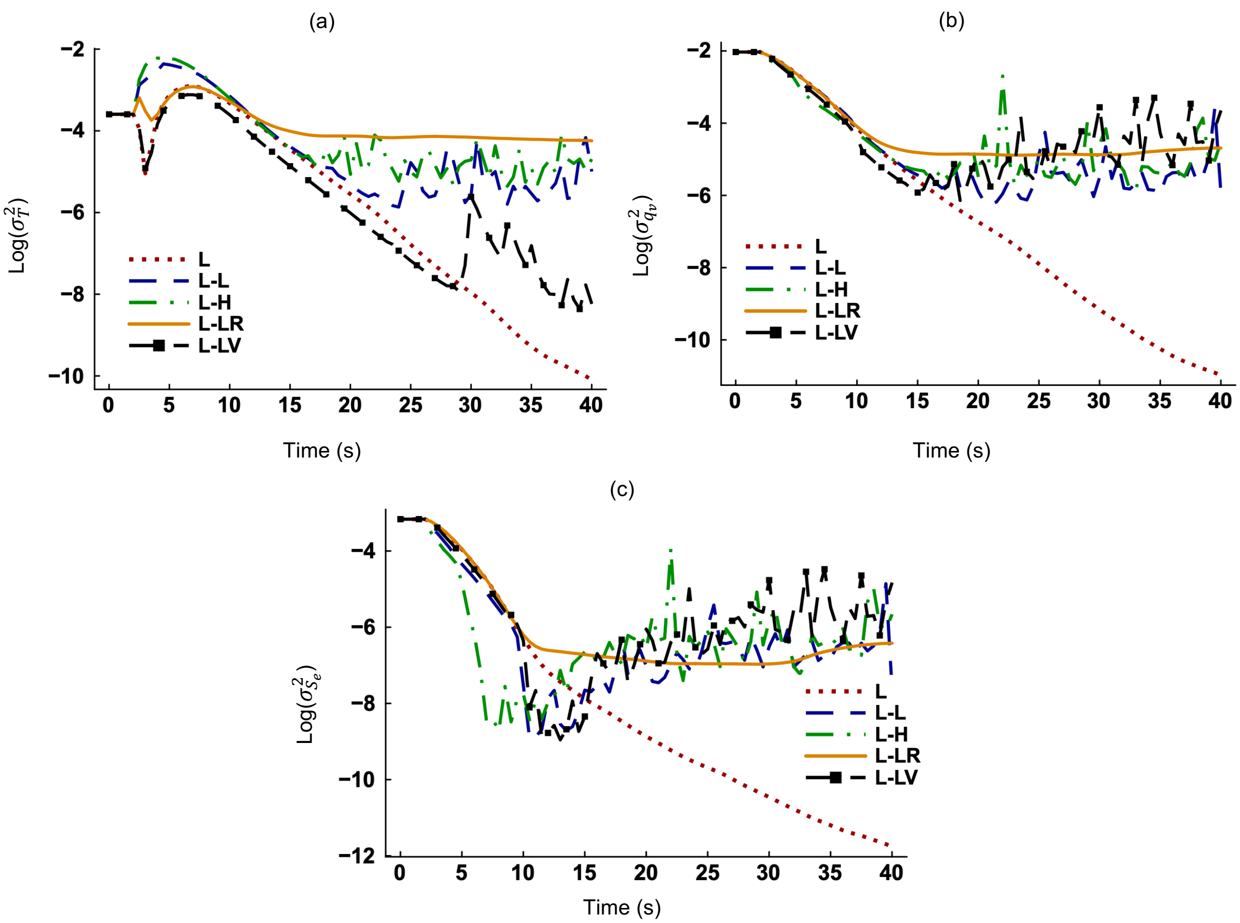

FIG. 2 Time evolution of the variances of (a) temperature, (b) vapor mixing ratio, and (c) environmental supersaturation in Scenario I.

External thermodynamic forcing produces fluctuations independent of those generated by momentum forcing. To understand this, we plot the two-point autocorrelation coefficient $R_{q_v'q_v'}$ against separation distance in Figs. 3(a)-(b). The $R_{q_v'q_v'}$ is defined as[32]

$$R_{q_v'q_v'}(\boldsymbol{r}) = \frac{\langle q_v'(\boldsymbol{x},t)q_v'(\boldsymbol{x}+\boldsymbol{r},t)\rangle}{\sqrt{\langle\left(q_v'(\boldsymbol{x},t)\right)^2\rangle\langle\left(q_v'(\boldsymbol{x},t)\right)^2\rangle}}\,, \qquad \boldsymbol{r} = \Delta x\boldsymbol{i} + \Delta y\boldsymbol{j} + \Delta z\boldsymbol{k}\,. \tag{23}$$

where $\boldsymbol{r}$ is the separation distance vector and q_v' is the fluctuating vapor mixing ratio. We see in Fig. 3(a) that $R_{q_v'q_v'}$ is unity when separation distance is zero and decays gradually for Case L. The complete decay of $R_{q_v'q_v'}$ happens at a shorter distance when thermodynamic forcing acts in random directions (Case L-LR) compared to when it is absent (L). Moreover, the decay takes place almost instantaneously for Cases L-L, L-H, and L-LV. For Case L, fluctuations in the T and q_v are induced by momentum forcing only. For the thermodynamically forced cases, both thermodynamic and momentum forcings contribute to the temperature and vapor fluctuations. As temperature, vapor, and momentum forcings produce fluctuations independently, their combined fluctuations decorrelate within a shorter distance. Figure 3(b) supports this finding. For Cases L and H, the only source of fluctuations in the T and q_v is momentum forcing. The correlation lengths for these two cases are almost identical (~ 0.16 m) while those for Cases L-L and H-L are ~ 0.05 m and ~ 0.07 m, respectively. The correlation length is shorter if multiple, independent sources of fluctuations are present (Cases L-L and H-L) as opposed to a single source (Cases L and H) because of destructive interference.[33] Thus, thermodynamic forcing produces spatially uncorrelated fluctuations. An implication of uncorrelated fluctuations is that neighboring particles are more likely to experience different growth histories.

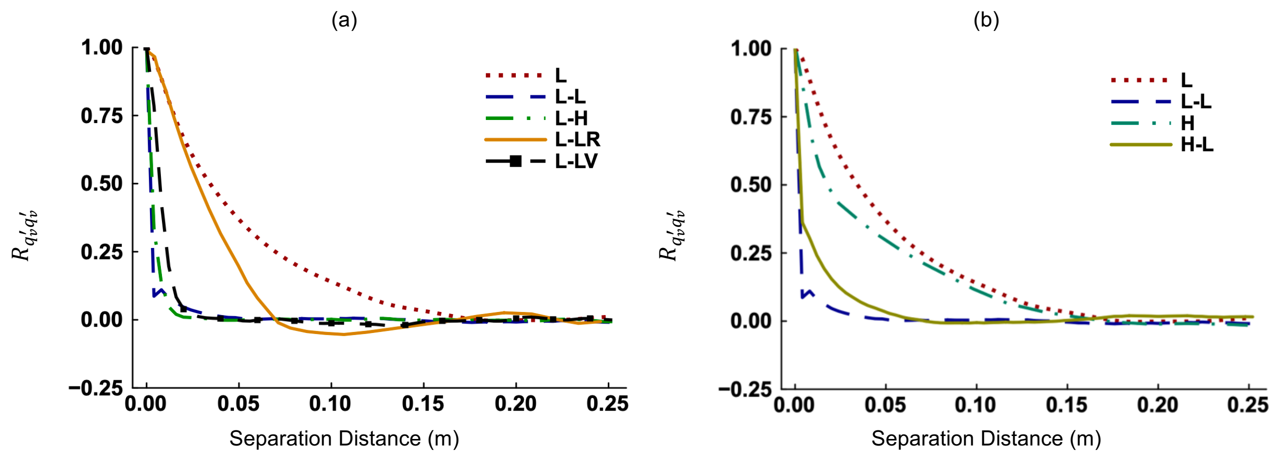

FIG. 3 Spatial correlations between fluctuations in the vapor mixing ratio for (a) Cases L, L-L, L-H, L-LR, and L-LV and (b) Cases L, L-L, H, H-L. The results represent Scenario I at 30 s.

Figures 4(a)-(c) show the temporal evolution of mean thermodynamic fields in Scenario I. We find that $\langle S_e \rangle$ is 0.025 initially and decreases to zero for all cases (Fig. 4(c)). When $\langle S_e \rangle \sim 0$, saturation is reached and particles do not grow or shrink further unless the S_e' is substantial. This indicates a state of equilibrium as new particles or water vapor are not allowed to enter or leave the model. The temporal increase in the $\langle T \rangle$ (Fig. 4(a)) is accompanied by a decrease in the $\langle q_v \rangle$ (Fig. 4(b)) for all cases. During condensation, water vapor converts into liquid water and the $\langle q_v \rangle$ decreases. It appears that $\langle T \rangle$ and $\langle q_v \rangle$ begin to evolve only after particles start to grow (~2 s). This is because the latent heat (phase change) terms influence the evolution of $\langle T \rangle$ and $\langle q_v \rangle$ more strongly than the convection and diffusion terms do. More details in this regard are provided in Appendix B. The evolution of $\langle T \rangle$ and $\langle q_v \rangle$ appears to stop at 6 s, 10 s, 10 s, and 40 s for Cases L-H, L-L, L-LV, and L, respectively when $\langle S_e \rangle \sim 0$ for the respective cases. A total equilibrium is not reached for Case L-LR by 40 s. We see that the $\langle T \rangle$ increases more for Cases L-L and L-H but less for Cases L-LR and L-LV compared to the unforced case (L). Thus, the impact of external forcing on mean thermodynamic fields depends on the field and direction it is applied to. Again, the $\langle q_v \rangle$ decreases less for Cases L-L and L-H but more for

Cases L-LR and L-LV compared to Case L. The magnitude of thermodynamic forcing is also important as shown by the different results for Cases L-L and L-H (Figs. 4(a)-(c)). We notice that $\langle T \rangle$ increases least but $\langle q_v \rangle$ decreases most if the vapor field is forced but the temperature field in unforced (Case L-LV). Hence, forcing of one thermodynamic field (vapor) influences the other (temperature) in Scenario I.

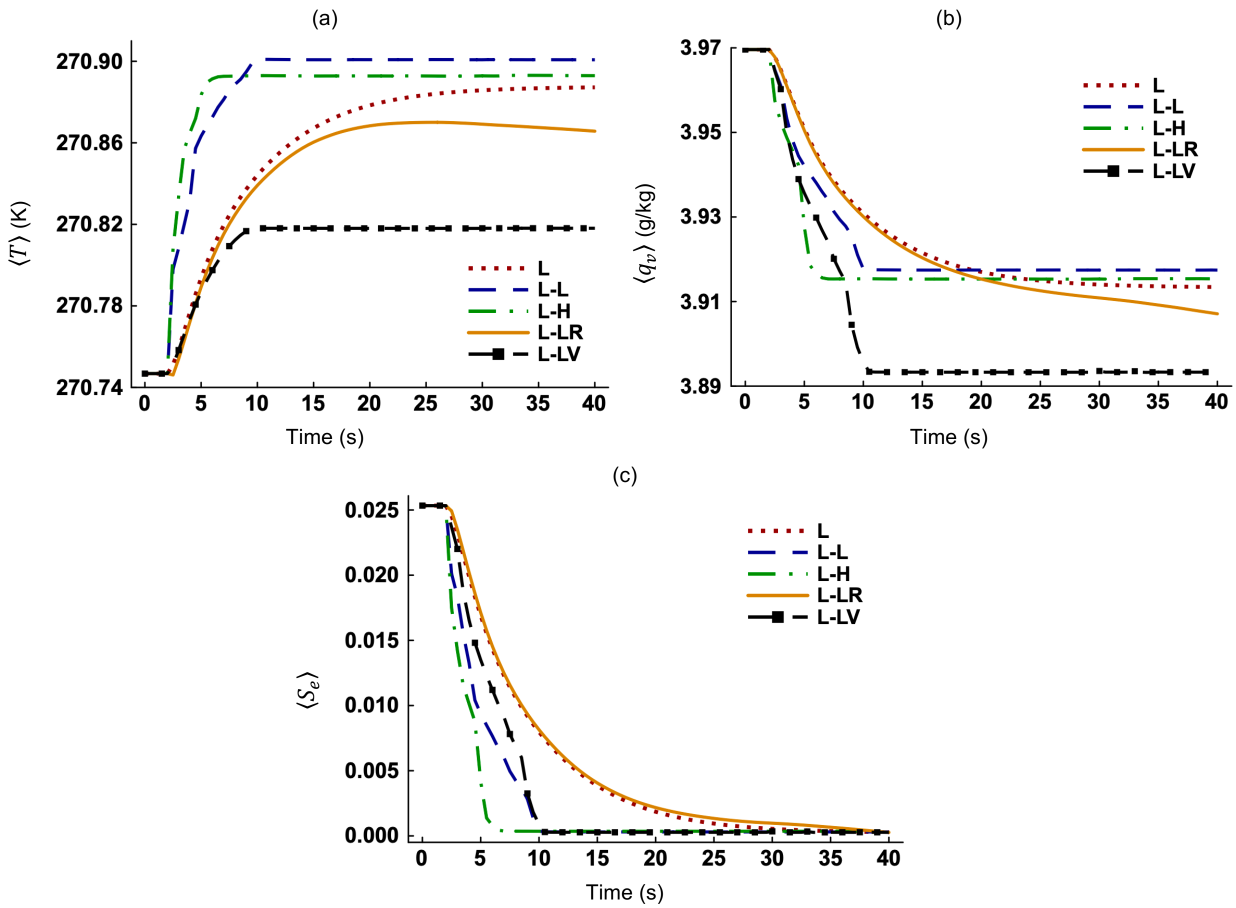

FIG. 4 Time evolution of the (a) mean temperature ($\langle T \rangle$), (b) mean vapor mixing ratio ($\langle q_v \rangle$), and (c) mean environmental supersaturation ($\langle S_e \rangle$) in Scenario I.

2. Evaporative decay (Scenario II)

Scenario II focuses on evaporative decay and cloud droplet deactivation. The temporal evolution of the relative dispersions of thermodynamic fields is plotted in Figs. 5(a)-(c) which show the decay of fluctuating thermodynamic fields in absence of external forcing. Again, statistical stationarity is reached with inclusion of forcing. The $\sigma_T/\langle T \rangle$ and $\sigma_{q_v}/\langle q_v \rangle$ are $6.8e-6$ and $1.07e-3$, respectively, for Case L at 30 s. The corresponding values for Case L-L are $6.1e-5$ and $1.62e-2$. Although forcing increases fluctuations in the temperature and vapor mixing ratio, the σ_T and σ_{q_v} remain much smaller than the respective means in our model. This is consistent with the standard deviations measured in natural clouds.[34] The $\sigma_{S_e}/\langle S_e \rangle$ is $3.1e-2$ for Case L but 27.78 for Case L-L at 30 s. Hence, the σ_{S_e} is larger than the $\langle S_e \rangle$ with thermodynamic forcing. The same has been reported by Paoli et al.[12] who applied Fourier forcing in the temperature and vapor mixing ratio fields. The $\langle S_e \rangle$ is much smaller than the $\langle T \rangle$ and $\langle q_v \rangle$, which makes fluctuations in the supersaturation more pronounced relative to the mean.

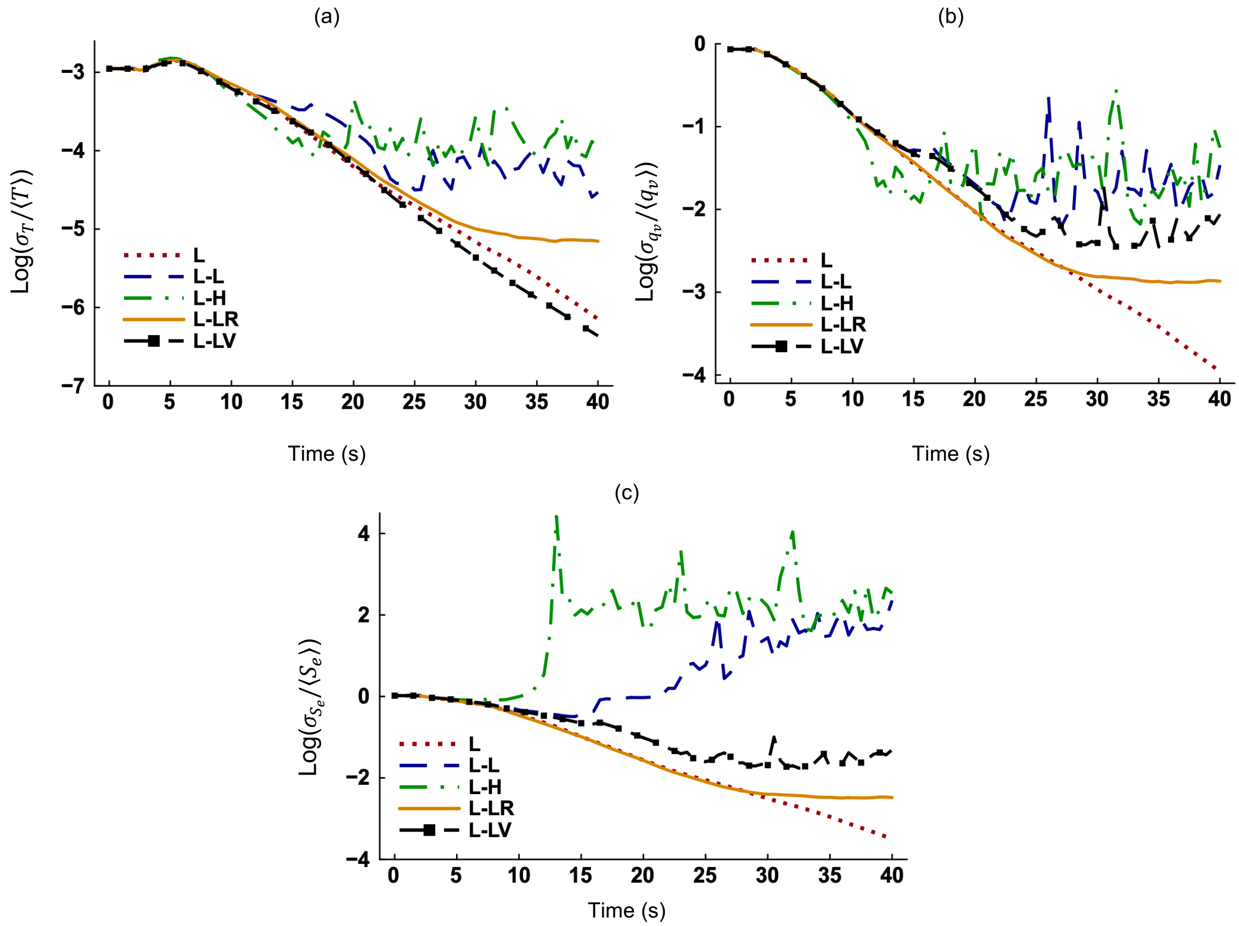

FIG. 5 Time evolution of the relative dispersions of (a) temperature, (b) vapor mixing ratio, and (c) environmental supersaturation in Scenario II.

Figure 6(a) shows the PDFs of T'/σ_T, which is the fluctuating temperature normalized by standard deviation, at 30 s in Scenario II. Figure 6(b) represents the same for vapor mixing ratio. It appears that the normalized PDFs in both figures are more spread out for Case L-LR compared to the other cases. We quantify the skewness (γ) and kurtosis (ω) of the probability distributions as[35]

$$\gamma = \frac{\langle (\varphi')^3 \rangle}{\langle (\varphi')^2 \rangle^{3/2}}, \quad \omega = \frac{\langle (\varphi')^4 \rangle}{\langle (\varphi')^2 \rangle^{4/2}}, \quad \varphi = \frac{\theta'}{\sigma_\theta}. \tag{24}$$

The γ of temperature PDFs in Fig. 6(a) are 0.099, 0.039, -0.0006, -2.35, and -0.36 for Cases L, L-L, L-H, L-LR, and L-LV, respectively, while the corresponding ω are 2.41, 5.07, 1.76, 65.33, and 2.48. The γ of vapor PDFs in Fig. 6(b) are -0.14, 0.0018, 0.0006, -2.28, and -0.15, respectively, while the corresponding ω are 2.39, 3.68, 2.43, 117.5, and 7.69. We see that the PDFs can be negatively or positively skewed depending on the details of forcing. Skewness is 0 and kurtosis is 3 for a perfectly Gaussian distribution. Based on the kurtosis values, none of the distributions in Figs. 6(a)-(b) are Gaussian. The PDFs of thermodynamic fluctuations have been reported to be non-Gaussian in previous studies of cloud turbulence.[36] For Case L-LV, the vapor PDF has a sharp peak (Fig. 6(b)) but the peak of temperature PDF is somewhat rounded (Fig. 6(a)). This is expected because ω $(= 7.69) > 3$ for the former means a leptokurtic distribution and a sharp peak while ω $(= 2.48) < 3$ for the latter suggests a platykurtic distribution and a rounded peak. The deviation from Gaussianity[37] is strongly noticeable with random thermodynamic forcing (Case L-LR) that correspond to kurtosis values of 65.33 and 117.5 for the PDFs of T'/σ_T and q_v'/σ_{q_v}, respectively. Moreover, extreme fluctuations exist for this case with maximum T' being $30\sigma_T$ (Fig. 6(a)) and maximum q_v' reaching $40\sigma_{q_v}$ (Fig. 6(b)). On the other hand, maximum T' is $\sim 3\sigma_T$ for Case L-H and is $\sim 8\sigma_T$ for Case L-L while maximum q_v' is $\sim 4\sigma_{q_v}$ for Case L-L and is $\sim 10\sigma_{q_v}$ for Case L-LV. Anderson et al.[38] reported similar ranges for thermodynamic fluctuations in their experimental investigation of cloud processes. They quantified that maximum T' is $\sim 4\sigma_T$ and maximum q_v' is $\sim 6\sigma_{q_v}$ in the updraft region while maximum T' is $\sim 5\sigma_T$ and maximum q_v' is $\sim 4\sigma_{q_v}$ in the downdraft region.

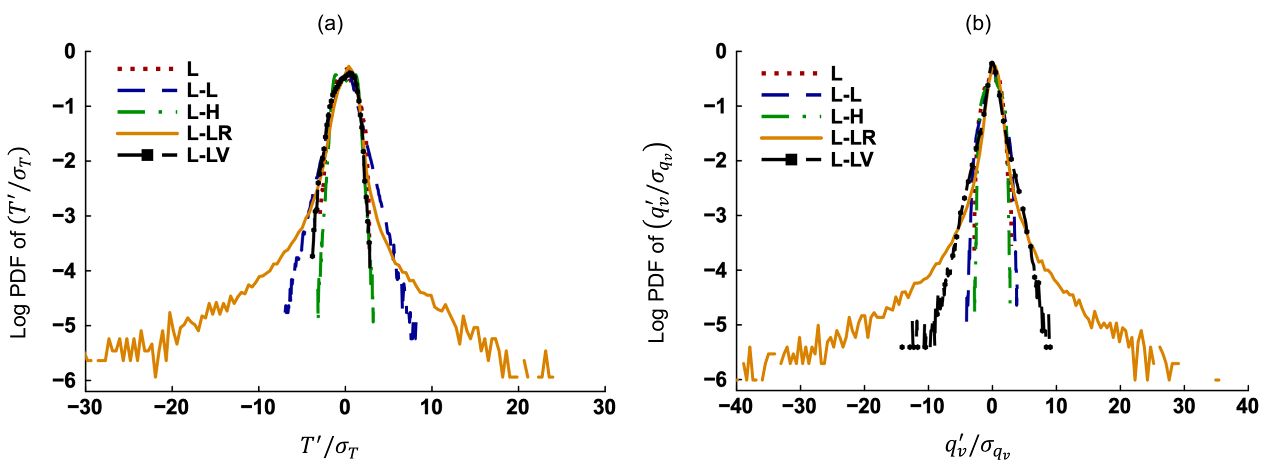

FIG. 6 Log PDFs of the (a) fluctuating temperature normalized by standard deviation (T'/σ_T), and (b) fluctuating vapor mixing ratio normalized by standard deviation (q_v'/σ_{q_v}) at 30 s in Scenario II.

The time evolution of mean thermodynamic fields for Scenario II is shown in Figs. 7(a)-(c). During evaporation, liquid water converts into water vapor, increasing the $\langle q_v \rangle$ and $\langle S_e \rangle$. Meanwhile, the $\langle T \rangle$ decreases in a manifestation of evaporative cooling. Without thermodynamic forcing, an increase in the $\langle T \rangle$ corresponds to a decrease in the $\langle q_v \rangle$

because phase change terms have opposite signs in Eqs. (5) and (6). That order is apparently broken with inclusion of thermodynamic forcing as Figs. 7(a)-(b) show that $\langle T \rangle$ decreases most for Case L-L but $\langle q_v \rangle$ does not increase most for this case. The temperature and vapor forcing terms are independent of each other and influence the evolution of $\langle T \rangle$ and $\langle q_v \rangle$ separately. We also see that thermodynamic forcing drives the model from subsaturation to saturation ($\langle S_e \rangle \sim 0$) for Cases L-L and L-H (Fig. 7(c)). Between these two cases, a higher magnitude of forcing (L-H) leads to equilibrium faster. Random thermodynamic forcing (Case L-LR) has negligible impact on the evolution of $\langle T \rangle$, $\langle q_v \rangle$, and $\langle S_e \rangle$ compared to Case L in Scenario II. Again, vapor forcing affects the $\langle q_v \rangle$ and $\langle S_e \rangle$ but no temperature forcing leaves the $\langle T \rangle$ unaffected for Case L-LV relative to Case L. When cloud droplets fully evaporate, there is no feedback from microphysics to thermodynamics and thus the $\langle T \rangle$, $\langle q_v \rangle$, and $\langle S_e \rangle$ become steady before saturation is reached (Cases L, L-LR, and L-LV).

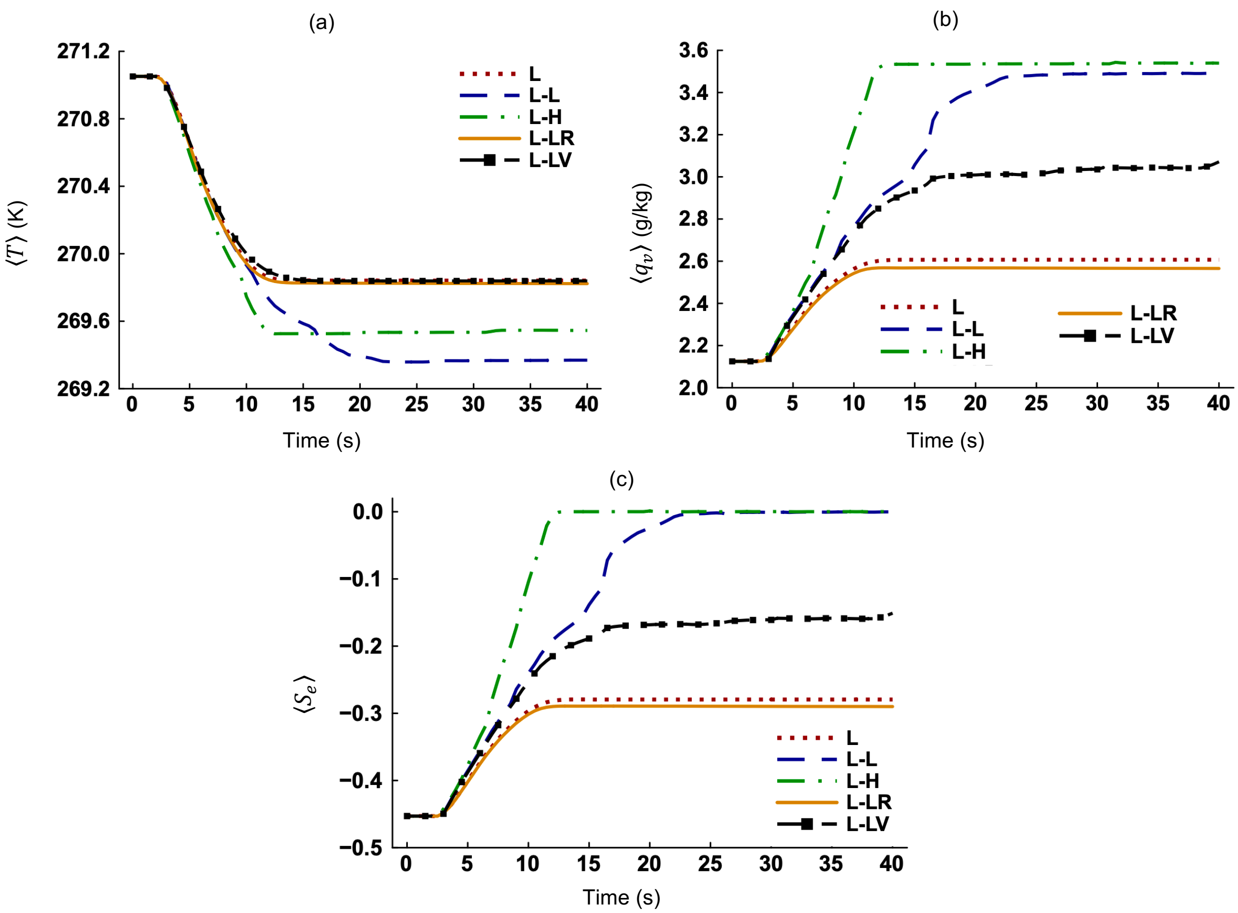

FIG. 7 Time evolution of the (a) mean temperature ($\langle T \rangle$), (b) mean vapor mixing ratio ($\langle q_v \rangle$), and (c) mean environmental supersaturation ($\langle S_e \rangle$) in Scenario II.

B. The impact of thermodynamic forcing on microphysics

1. Condensation and activation (Scenario I)

Figure 8(a) suggests that the mean radii tend to asymptote after exponential increase at early times. This is due to particle growth (Eq. (13)) slowing down over time as the r increases and S_e decreases (Fig. 4(c)). Condensational growth is marginally higher if thermodynamic forcing acts in random directions (Case L-LR) compared to when it is absent (Case L). On the other hand, particle growth is significantly lower for the other forced cases (L-L, L-H, and L-LV). When thermodynamic forcing acts in the direction of phase change, the supersaturated model reaches saturation faster (Fig. 4(c)). That leaves less time available for the completion of phase change and reduces particle growth (condensation). The standard deviation of particle radii (σ_r) increases very fast between 2 s and 5 s for all cases (Fig. 8(b)) which is consistent with exponential early growth. Then σ_r decreases slightly for Cases L and L-LR but increases very slowly for Cases L-L, L-H, and L-LV. The σ_r profiles converge[39] for Cases L and L-H at long time.

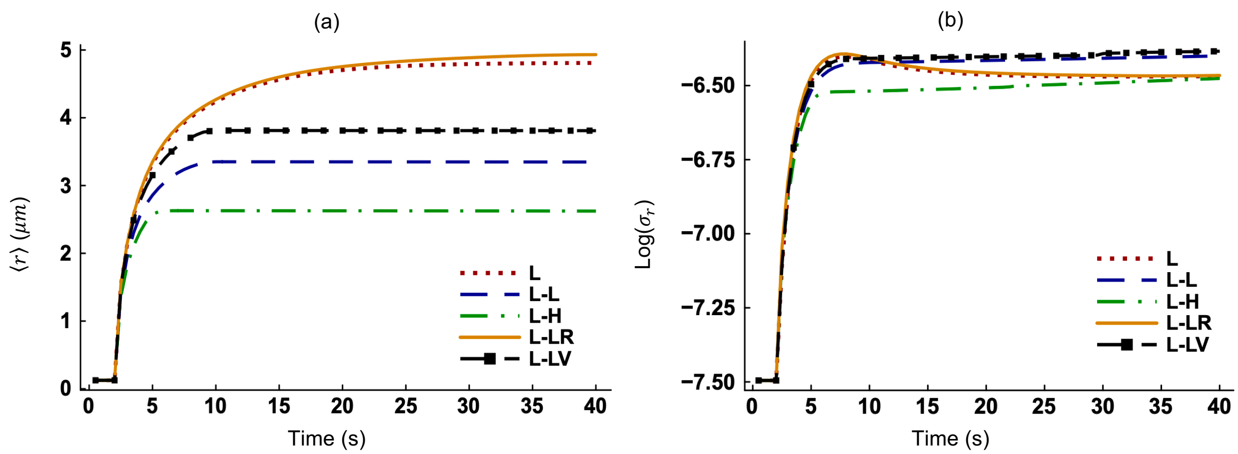

FIG. 8 Time evolution of the (a) mean radius ($\langle r \rangle$) of all particles and (b) standard deviation of their radii (σ_r) in Scenario I.

The PDFs of the radii of cloud droplets and aerosol particles (f_r) are shown in Figs. 9(a)-(d) at 30 s in Scenario I. Only dry aerosols exist at 0 s as prescribed by the initial condition. By 30 s, a portion of them activates into cloud droplets. Activation is determined by the critical radius (r_c) which has its own distribution since the dry radii (r_d) are lognormally distributed. The activation barrier varies particle to particle. Some particles are still classified as aerosol particles ($r < r_c$) despite having a larger radii than some cloud droplets ($r > r_c$). Consequently, aerosol particles and cloud droplets coexist as shown in Figs. 9(a)-(d). Their radius PDFs are skewed to the left and most cloud droplets and aerosol particles are larger than the respective mean values. While the PDFs have identical spreads and shapes for Cases L (Fig. 9(a)) and L-LR (Fig. 9(d)), the minimum, mean, and maximum cloud droplet radii are larger by around 0.2 μm for the latter. Therefore, random thermodynamic forcing increases the radii of cloud droplets uniformly during condensation. The f_r of cloud droplets and aerosol particles have a larger spread for Case L-L (Fig. 9(b)) compared to Case L and so does the f_r of aerosol particles for Case L-H (Fig. 9(c)). Thus, thermodynamic forcing broadens the size distributions of particles in our model. Satio et al.[14] applied temperature and vapor forcings in a $1.024^3\ m^3$ DNS domain to produce broader droplet size distributions. We also show that thermodynamic forcing is useful to obtain broader f_r which are more common in experiments.[40] Figures 9(a)-(d) imply that the expected (mean) radii are smaller for Cases L-L and L-H compared to those for Case L, consistent with Fig. 8(a).

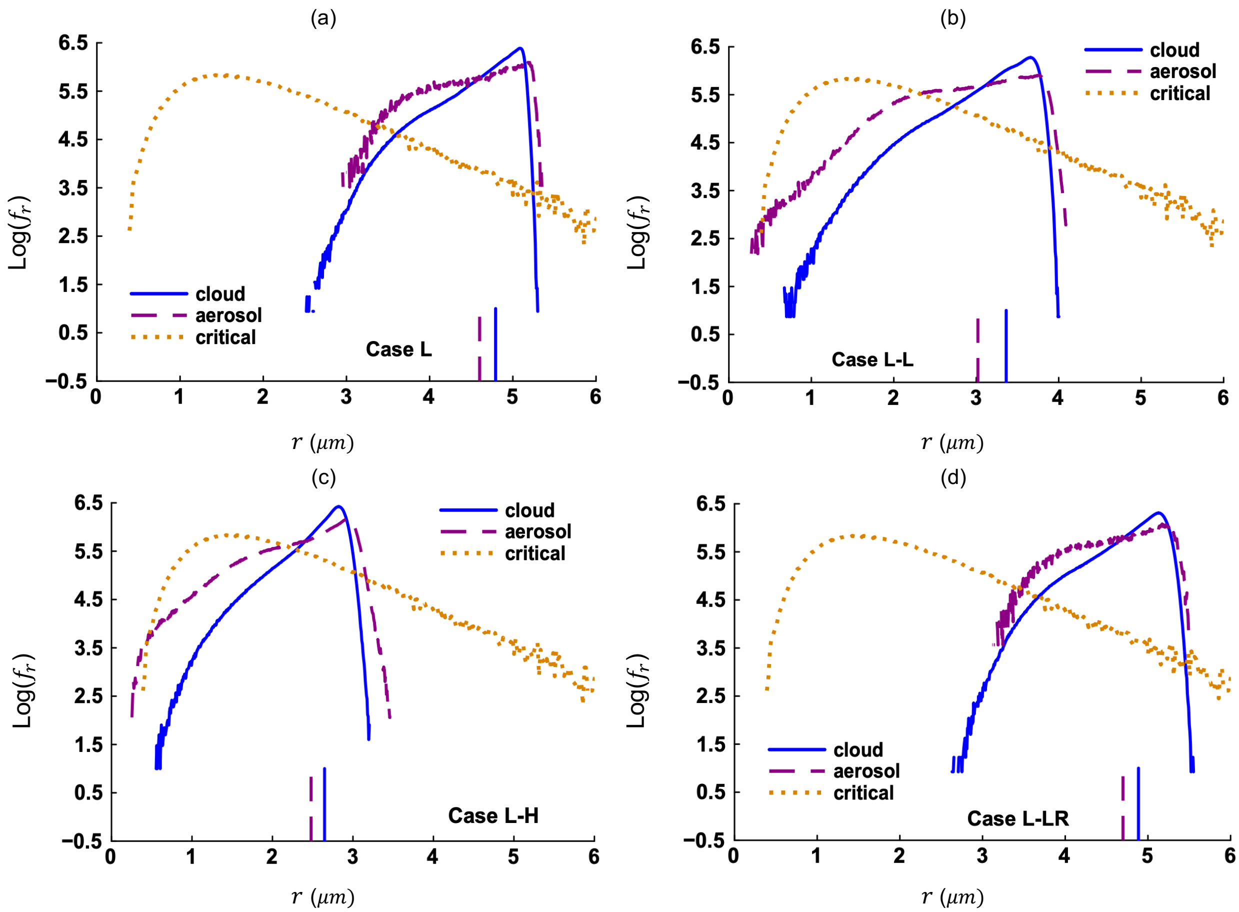

FIG. 9 Log PDFs of particle radii at 30 s for (a) Case L, (b) Case L-L, (c) Case L-H, and (d) Case L-LR in Scenario I. The vertical lines show the expected (mean) radii from respective PDFs. The critical radius (r_c) distribution is also shown.

2. Evaporation and deactivation (Scenario II)

Cloud droplets have a monodisperse size distribution of 10 μm initially in Scenario II and experience evaporative decay afterwards. Figure 10(a) indicates that the mean radii approach zero for Cases L, L-LR, and L-LV as all liquid water converts into water vapor. But the $\langle r \rangle$ does not become exactly zero because dry aerosols remain in the domain even though evaporation is complete. We note that thermodynamic forcing slows down phase change (evaporative decay)[41] for Cases L-L and L-H (Fig. 10(a)). Interestingly, the $\langle r \rangle$ increases a little after decreasing almost linearly for these two forced cases. The increase in Fig. 10(a) begins at the same time when the $\langle S_e \rangle$ becomes zero (Fig. 7(c)) for the respective cases. With $\langle S_e \rangle \sim 0$, fluctuations in the S_e influence particle growth considerably. Negative fluctuations cannot reduce the r further if it already equals to the r_d while positive fluctuations can always make the r bigger. This leads to a net increase in the $\langle r \rangle$. Figure 10(b) suggests that the σ_r increases, reaches a peak, and then decreases for all cases in Scenario II. The decrease is more noticeable for Cases L, L-LR, and L-LV at late times when cloud droplets have completely evaporated (Fig. 10(a)). Whereas partial evaporation for Cases L-L and L-H causes cloud droplets to exist in various sizes and supports a higher standard deviation (Fig. 10(b)).

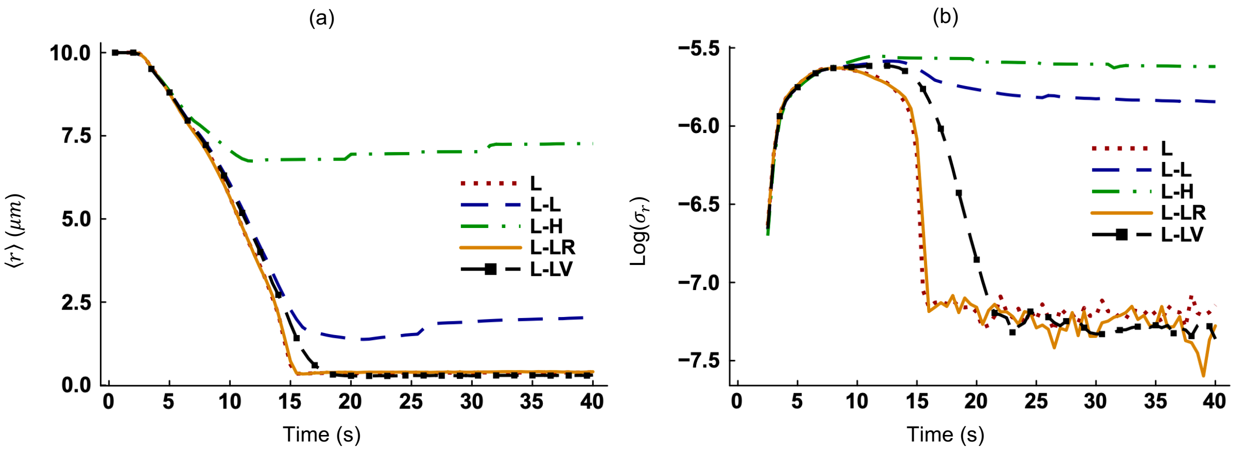

FIG. 10 Time evolution of the (a) mean radius ($\langle r \rangle$) of all particles and (b) standard deviation of their radii (σ_r) in Scenario II.

The radius PDFs (f_r) in Figs. 11(a)-(d) show that no cloud droplets exist at 30 s for Cases L and L-LR due to complete evaporation. The remaining dry aerosols in both figures follow a lognormal distribution. On the other hand, thermodynamic forcing suppresses evaporation for Cases L-L and L-H and their f_r are broader. Although most cloud droplets shrink, some of them grow close to 12 μm (Fig. 11(c)) if the magnitude of thermodynamic forcing is high (Case L-H). This agrees with the observation that thermodynamic and microphysical fluctuations are less correlated if thermodynamic forcing is present (Figs. 3(a)-(b)) and a locally supersaturated condition is created while the overall environment is subsaturated. Thus, some droplets evolve differently (grow) than others do (shrink). The f_r of aerosol particles is bimodal for Cases L-L and L-H, with one peak for smaller particles and another for larger particles (Figs. 11(b)-(c)). Bimodal size distribution of aerosol particles[42] has been observed in atmospheric clouds.

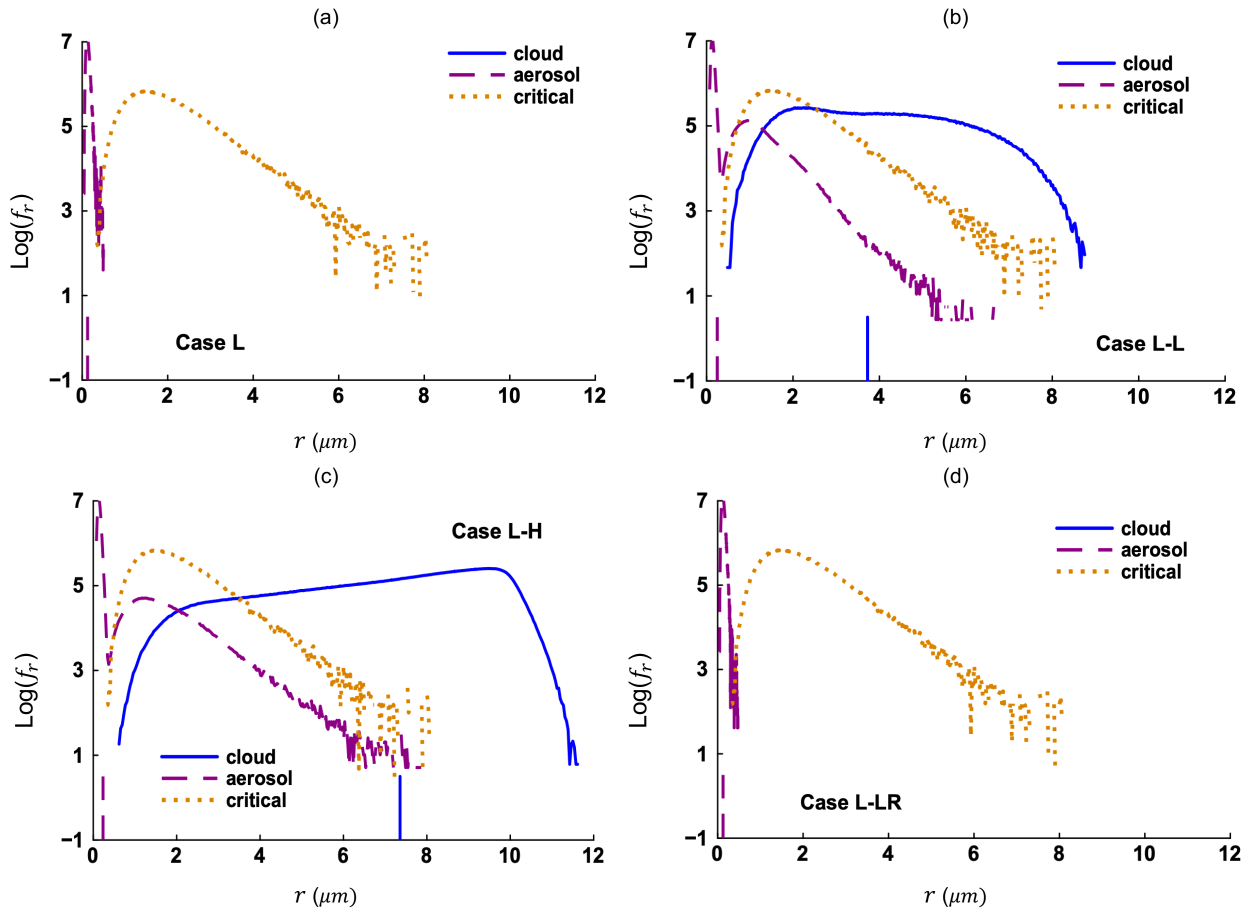

FIG. 11 Log PDFs of particle radii at 30 s for (a) Case L, (b) Case L-L, (c) Case L-H, and (d) Case L-LR in Scenario II. The vertical lines show the expected (mean) radii from respective PDFs. The critical radius (r_c) distribution is also shown.

C. The impact of external forcing on turbulent mixing

The entrainment and mixing of clear air with cloudy air is simulated in this study. In this section, we discuss the combined effects of momentum and thermodynamic forcings on entrainment and mixing by comparing the results for Case L (low momentum forcing, no thermodynamic forcing), Case L-L (low momentum forcing, low thermodynamic forcing), Case H (high momentum forcing, no thermodynamic forcing), and Case H-L (high momentum forcing, low thermodynamic forcing). Figure 12(a) shows that the TKE dissipation rates are identical for Cases L and L-L, and for Cases H and H-L. So, flow turbulence is affected by momentum forcing only with no contribution of thermodynamic forcing. The relative dispersion of supersaturation ($\sigma_{S_e}/\langle S_e \rangle$) decays faster for Case H. Initially, the thermodynamic fields vary only at the cloud/clear-air interface. Afterwards, turbulent mixing spreads thermodynamic fluctuations to all over the domain. This process is accelerated by a higher level of flow turbulence (momentum forcing). On the contrary, thermodynamic forcing counteracts the decay (Fig. 12(a)). The profiles of $\sigma_{S_e}/\langle S_e \rangle$ are almost identical for

Cases L-L and H-L (Fig. 12(b)) because thermodynamic forcing overrides the homogenization caused by momentum forcing. The $\sigma_r/\langle r \rangle$ are similar for these two forced cases in Scenario I at long time (Fig. 12(c)) while strong mixing reduces the $\sigma_r/\langle r \rangle$ for Case H compared to Case L. During evaporation (Fig. 12(d)), the $\sigma_r/\langle r \rangle$ increases first for all cases as monodisperse cloud droplets are released in a turbulent environment and grow at different rates. The $\sigma_r/\langle r \rangle$ decreases after reaching a peak for Cases L and H as turbulent mixing leads to complete evaporation and reduces the standard deviation. The decrease in the $\sigma_r/\langle r \rangle$ is apparently prevented by thermodynamic forcing (Cases L-L and H-L).

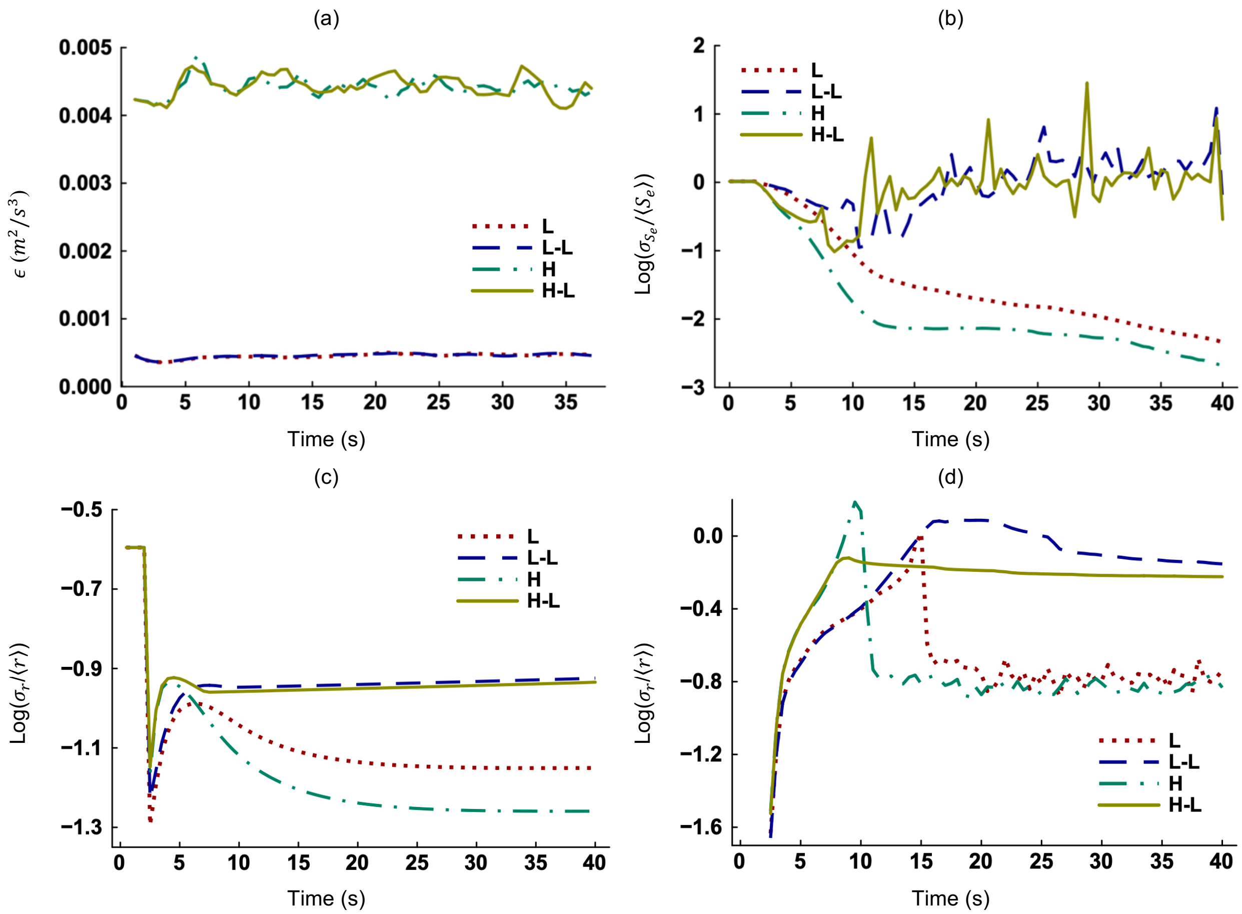

FIG. 12 Time evolution of the (a) TKE dissipation rate (ϵ), (b) relative dispersion of environmental supersaturation ($\sigma_{S_e}/\langle S_e \rangle$), and (c) relative dispersion of particle radii ($\sigma_r/\langle r \rangle$) in Scenario I. The subplot (d) represents the $\sigma_r/\langle r \rangle$ in Scenario II.

The Damkohler number (Da) is defined as the ratio of the turbulent mixing timescale (τ_t) to the phase change (condensation or evaporation) timescale (τ_c):[43]

$$Da = \frac{\tau_t}{\tau_c} = \frac{\left(\frac{l^2}{\epsilon}\right)^{1/3}}{\frac{r^2}{|G(S_e - S_k)|}}, \qquad l = \int_0^\infty R_{w'w'}(\boldsymbol{r})dr\,. \tag{25}$$

where l is the integral length scale[35] and the $R_{w'w'}$ is calculated using Eq. (23) with w' being the fluctuating z-velocity. The Da is inversely proportional to the r^2 and thus decreases during condensation. The sharp decline between 2 s to 4 s (Fig. 13(a)) corresponds to the exponential growth at early times (Fig. 8(a)). Conversely, the Da increases during evaporation. We notice instantaneous spikes in Fig. 13(b) at 8 s and 14 s for Cases H and L, respectively. This happens when the r shrinks extremely fast as particles get smaller (Eq. (13)). The decrease and increase in the Da follows the order: Case H > Case L for both scenarios if thermodynamic forcing is absent. When present (Cases L-L and H-L), the Da changes slowly compared to Cases L and H. It is evident that thermodynamic forcing keeps the Damkohler number close to unity ($log(Da)$~0) at long time (Figs. 13(a)-(b)). Consequently, a balance is maintained between turbulent mixing and microscale phase change.

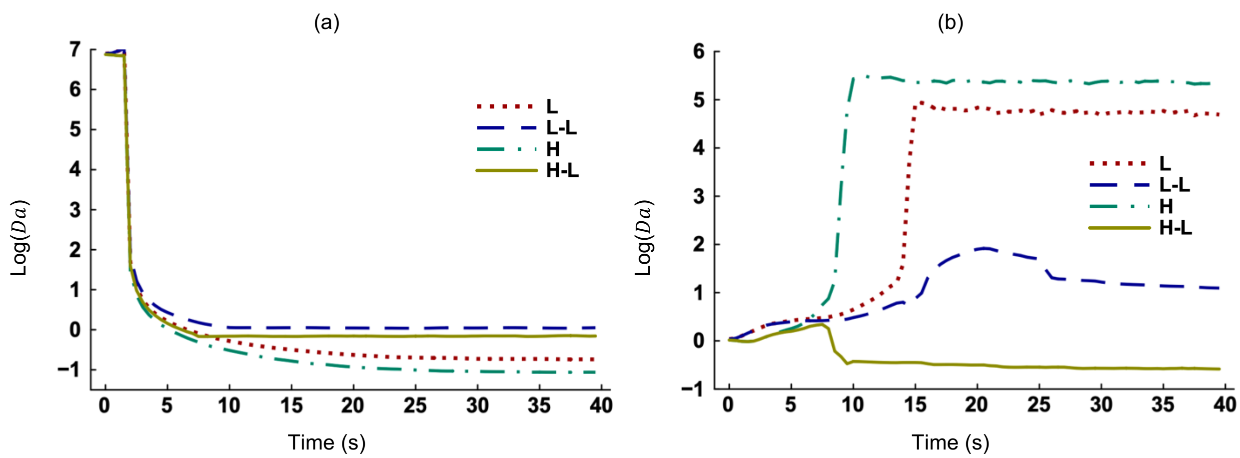

FIG. 13 Time evolution of the Damkohler number (Da) in (a) Scenario I and (b) Scenario II.

V. CONCLUDING REMARKS

A three-dimensional particle-based direct numerical simulation (DNS) model is employed with external forcing in the momentum and thermodynamic fields. The Fourier forcing method is implemented. A realistic initial setup is used to study the growth and decay of cloud droplets. The key findings from our simulations are itemized below.

- The mean and fluctuating thermodynamic fields evolve from nonzero initial conditions. External forcing leads thermodynamic fluctuations to statistical stationarity (Figs. 2 and 5). Unlike the mean velocity fields, the mean temperature and water vapor are not zero in our model. Hence, external forcing affects the evolution of mean thermodynamic fields (Figs. 4 and 7).

- Temperature, vapor, and momentum forcings act as independent sources of thermodynamic fluctuations and causes the latter to decorrelate within a shorter distance (Figs. 3(a)-(b)). Particles are more likely to experience different growth histories (Fig. 11(c)) if forcing is strong and fluctuations are uncorrelated.
- Forcing of one thermodynamic field (vapor) influences the other (temperature) during condensation (Fig. 4). But vapor forcing does not affect the temperature field during evaporation (Fig. 7). Random thermodynamic forcing has negligible impact on the evolution of $\langle S_e \rangle$ relative to the unforced case in both scenarios (Figs. 4(c) and 7(c)).
- The PDFs of fluctuating temperature and vapor fields have high kurtosis and are non-Gaussian with random thermodynamic forcing (Fig. 6). Extreme thermodynamic fluctuations are observed for this case. For the other forced cases, the ranges of T' and q_v' match those reported by Anderson et al.[38]
- When thermodynamic forcing acts in the direction of phase change, the model reaches saturation ($\langle S_e \rangle \sim 0$) faster (Figs. 4(c) and 7(c)). That leaves less time available for the completion of phase change and reduces the rates of condensation (Fig. 8(a)) and evaporation (Fig. 10(a)).
- Satio et al.[14] pointed out that the particle size distributions (f_r) produced by DNS are much narrower than those obtained from experiments and thermodynamic forcing with sufficient domain size can broaden the f_r. We report the same during condensational growth (Figs. 9(a)-(c)).
- If the magnitude of thermodynamic forcing is high, a locally supersaturated condition is created although the overall environment is subsaturated. This allows some droplets to grow while most others shrink (Fig. 11(c)). A bimodal size distribution is observed for aerosol particles during partial evaporation (Figs. 11(b)-(c)).
- In absence of thermodynamic forcing, a higher level of momentum forcing (velocity turbulence) homogenizes the thermodynamic fields through enhanced mixing (Fig. 12(b)). However, thermodynamic forcing counteracts turbulent homogenization and prevents the decay of the relative dispersions of particle radii (Figs. 12(c)-(d)).
- The Damkohler number (Da) changes slowly if thermodynamic forcing is present and a balance is maintained between turbulent mixing and microscale phase change (Figs. 13(a)-(b)).
- The magnitude of thermodynamic fluctuations varies in natural clouds. We demonstrate that a DNS model with thermodynamic forcing can maintain fluctuations at desired levels and produce microphysical results that agree with physical observations.

ACKNOWLEDGEMENTS

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APPENDIX A: DIRECTIONALITY OF THERMODYNAMIC FORCING

We test Case L-L′ in which temperature and vapor forcings always act in the direction opposite to the change of latent heat (phase change). Figures 14(a)-(b) compares the time evolution of the $\langle S_e \rangle$ and $\langle r \rangle$ for Cases L and L-L′ in Scenario I. It shows that thermodynamic forcing tends to replenish the water vapor consumed during condensation and prevent the $\langle S_e \rangle$ from decaying to zero for Case L-L′ (Fig. 14(a)). Due to the availability of water vapor, cloud droplets keep growing (Fig. 14(b)). However, this type of unrestrained condensational growth is unlikely to happen in real clouds[44] because the amount of water vapor air can hold is limited by the temperature, pressure, and altitude. Hence, Case L-L′ does not produce realistic results in the context of our problem.

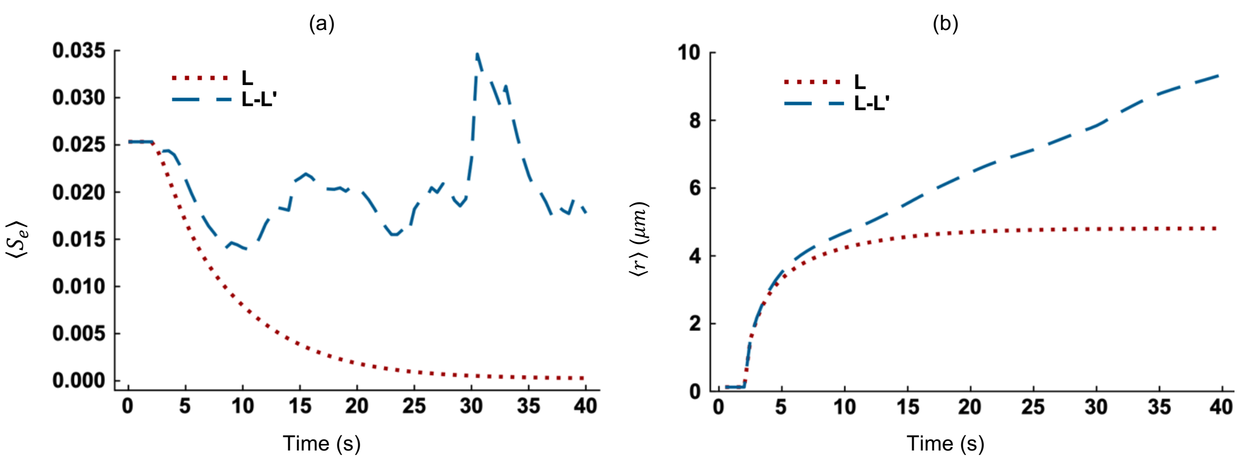

FIG. 14 Time evolution of the (a) mean environmental supersaturation ($\langle S_e \rangle$) and (b) mean particle radius ($\langle r \rangle$) in Scenario I.

APPENDIX B: INDIVIDUAL TERMS IN THE TRANSPORT EQUATIONS

By ignoring the convection and diffusion terms, Eq. (5) can be reduced to Eq. (B1) for Case L that assumes no external thermodynamic forcing ($F_T = 0$ and $F_{q_v} = 0$). We run a simulation by replacing Eq. (5) with Eq. (B1) and compare the results. Figure 15(a) suggests that exclusion of the convection and diffusion terms has minimum effect on the evolution of $\langle T \rangle$ which is mainly driven by the absorption of latent heat during evaporation. We compare the results for Eq. (B1) and Eq. (B2) in Fig. 15(b). Equation (B2) takes temperature forcing into account (Case L-L) but excludes convection and diffusion terms. It is clear that F_T contributes significantly to the evolution of $\langle T \rangle$ in addition to the latent heat term. This conclusion is also valid for vapor mixing ratio. The diffusion of temperature and vapor is negligible in the atmosphere. But convection, in the form of adiabatic ascent and descent, is often significant.[45] DNS models use a small domain size due to computational cost and cannot accommodate large-scale convection. Instead, external thermodynamic forcing can augment the effects of convection term in a microscale DNS model.

$$\frac{\partial T}{\partial t} \approx \frac{L_h}{c_p} C_d \,, \tag{B1}$$

$$\frac{\partial T}{\partial t} \approx \frac{L_h}{c_p} C_d + F_T \,. \tag{B2}$$

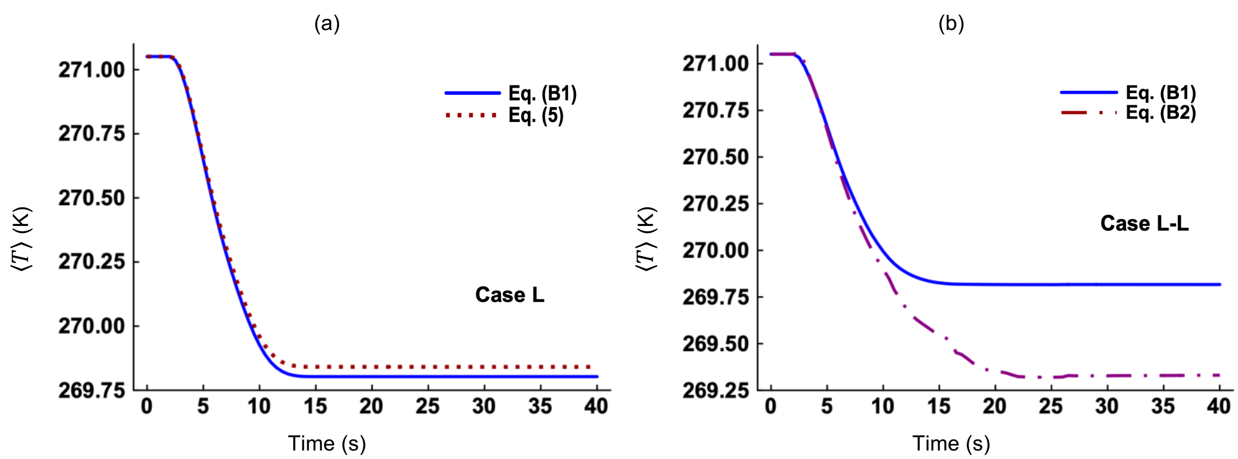

FIG. 15 Time evolution of the mean temperature in Scenario II using (a) Eqs. (5) and (B1) for Case L and (b) Eqs. (B1) and (B2) for Case L-L.

DATA AVAILABILITY

The data that support the findings of this work are available from the corresponding author upon reasonable request.

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